Hypoglycaemia

This guideline is for patients with a blood glucose concentration less than 3.5 mmol/litre. See the **'neonatal resuscitation'** guideline if the patient is a newborn baby.

- Administer 10-20 g of glucose PO provided the patient is sufficiently conscious to swallow safely.
- Administer glucose IV if the patient has a significantly altered level of consciousness or cannot swallow safely:
  - 100 ml of 10% glucose IV for an adult.
  - 2 ml/kg of 10% glucose IV for a child.
- Administer glucagon if unable to gain IV access:
  - 1 mg of glucagon IM for a patient aged five years and over.
  - 0.5 mg of glucagon IM for a child aged less than five years.
- Repeat the glucose measurement every ten minutes until the glucose concentration is consistently greater than 3.5 mmol/litre. Administer further doses of glucose if required, but do not repeat glucagon IM.

Referral and transport

- A patient may receive treatment for hypoglycaemia and be given a clear recommendation for treatment without transport providing the following criteria are met:
  - No red flags are present.
  - A competent adult can stay with the patient for the next four hours.
  - The patient eats a meal (preferably with complex carbohydrates).
  - The patient is given the hypoglycaemia information sheet, which is explained to them and the accompanying adult.
  - The patient is given a clear recommendation to have their treatment reviewed (for example by an appropriate healthcare professional or diabetes service personnel). If the patient is aged less than or equal to 18 years or has been recently diagnosed with diabetes, this review should occur within 24 hours if feasible.

Red flags (Should be seen in ED, usually via ambulance)

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- Overdose (including accidental) of insulin or oral hypoglycaemics.
- No known history of diabetes.
- Non-isolated episode.
- Taking an oral hypoglycaemic medicine and has hypoglycaemia without an obvious cause.
- Does not recover fully.
- Unable to mobilise normally.
- Unable to achieve a blood glucose concentration greater than 3.5 mmol/litre, ten minutes (or more) after glucagon or the last glucose administration.

Additional information

Causes of hypoglycaemia

- Hypoglycaemia usually occurs in a patient taking insulin or an oral hypoglycaemic medicine.
- Less common causes of hypoglycaemia include severe sepsis (particularly in young children), poisoning with medicines that lower blood glucose and liver failure.
- Some oral hypoglycaemic medicines are excreted primarily by the kidneys. If a patient taking an oral hypoglycaemic medicine develops hypoglycaemia without an obvious cause, unrecognised deterioration in kidney function may be a contributor.

Measuring blood glucose concentration

- Do not measure blood glucose concentration using the patient's glucose meter because the result may be inaccurate.
- Use capillary blood and not blood from other sources because glucose meters are calibrated to use capillary blood.
- Check the history of the patient's glucose meter whenever possible. In particular rule out recurrent hypoglycaemia.

Treating hypoglycaemia with oral glucose

- Oral glucose is best administered as a simple carbohydrate that is rapidly absorbed.

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- Examples include 10% glucose (100 ml contains 10 g of glucose), glucose tablets or gel, sugar dissolved in water, 'non-diet' jam or similar glucose containing spreads and 'non-diet' soft drinks.
- 10-20 g of glucose is approximately half a cup of 'non-diet' fruit juice or soft drink, one tablespoon of sugar, one tablespoon of honey or six squares of chocolate.
- Most oral glucose gels contain 10-20 g of glucose per dose and can be administered to all patients, regardless of age.
- Document oral glucose as the approximate number of grams of oral glucose administered, or a description of the food ingested.

- Glucose gel may be spread on the gums, tongue and inside of the cheeks of a baby or small child.
- Hypoglycaemia may occur several hours later as glucose is metabolised. This is why the instructions within the information sheet are important and why a meal must be eaten. The meal should preferably contain a complex carbohydrate such as brown rice, whole grain bread, porridge or beans.

Treating hypoglycaemia with glucagon

- Glucagon will be ineffective if the patient has depleted glycogen stores. Examples include if the patient does not have diabetes, is a young child, has sepsis, or has not eaten food for more than 12 hours. In this setting glucagon will not be harmful, but the focus must be on administration of glucose.

Hypoglycaemia in children aged less than five years

- Hypoglycaemia in children aged less than five years is most commonly due to:
  - An illness in combination with no food intake and depleted glycogen stores, or
  - Severe sepsis, or
  - Poisoning with oral hypoglycaemic medicines.
- Children not known to have diabetes are usually malnourished or glycogen depleted for another reason, and glucagon is unlikely to be effective.
- Although metabolic disorders can cause hypoglycaemia in young children, this is rare. Do not delay treatment of hypoglycaemia in young children pending diagnostic tests, other than measuring a capillary glucose concentration. Although some references advocate delaying treatment until a blood sample is taken for analysis of possible metabolic disorders, treatment always takes priority over blood tests in this setting.

Response and/or transport by helicopter

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- Response by helicopter may be indicated if it will arrive at scene at least 30 minutes sooner than a road-based Paramedic, and the patient is unconscious.

More Information

Version history

11 September 2023 1.0.4
Update: Multiple changes.